EVALUATION :

Choose the best answer

1. Electronic configuration of species M^{2+} is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6$ and its atomic weight is 56. The number of neutrons in the nucleus of species M is

   a) 26 b) 22 c) 30 d) 24

2. The energy of light of wavelength 45 nm is

   a) 6.67×10^{15}J b) 6.67×10^{11}J

   c) 4.42×10^{-18}J d) 4.42×10^{-15}J

3. The energies E_1 and E_2 of two radiations are 25 eV and 50 eV respectively. The relation between their wavelengths ie λ_1 and λ_2 will be

   a) $\frac{\lambda_1}{\lambda_2} = 1$ b) $\lambda_1 = 2\lambda_2$ c) $\lambda_1 = \sqrt{25 \times 50}\ \lambda_2$ d) $2\lambda_1 = \lambda_2$

4. Splitting of spectral lines in an electric field is called

   a) Zeeman effect b) Shielding effect

   c) Compton effect d) Stark effect

5. Based on equation $E = -2.178 \times 10^{-18}\ J \left(\frac{z^2}{n^2}\right)$, certain conclusions are written. Which of them is not correct ? (NEET)

   a) Equation can be used to calculate the change in energy when the electron changes orbit

   b) For n = 1, the electron has a more negative energy than it does for n = 6 which means that the electron is more loosely bound in the smallest allowed orbit

   c) The negative sign in equation simply means that the energy of electron bound to the nucleus is lower than it would be if the electrons were at the infinite distance from the nucleus.

   d) Larger the value of n, the larger is the orbit radius.

6. According to the Bohr Theory, which of the following transitions in the hydrogen atom will give rise to the least energetic photon ?

   a) n = 6 to n = 1 b) n = 5 to n = 4

   c) n = 5 to n = 3 d) n = 6 to n = 5

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